(FP1) 6. Vectors 专题作业

NO. 1

The position vectors of the points A, B, C, D are

$$\mathbf{i} + \mathbf{j} + 3\mathbf{k}, \quad 3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}, \quad -\mathbf{i} + 3\mathbf{k}, \quad m\mathbf{j} + 4\mathbf{k},$$

respectively, where m is a constant.

i. Show that the lines AB and CD are parallel when $m = \frac{3}{2}$. [1]

ii. Given that $m \neq \frac{3}{2}$, find the shortest distance between the lines AB and CD. [5]

iii. When $m = 2$, find the acute angle between the planes ABC and ABD, giving your answer in degrees. [6]

A level FURTHER MATHEMATICS 2018/Winter/Paper1_2/Q10

Mark scheme

Question	Answer	Marks	Guidance
10.i	$\overrightarrow{AB}=\begin{pmatrix}2\\3\\2\end{pmatrix}, \overrightarrow{CD}=\begin{pmatrix}1\\1.5\\1\end{pmatrix}=\frac{1}{2}\overrightarrow{AB}$	**B1**	Or shows if parallel, then m = 3/2
		1	
ii	$\overrightarrow{AB}=\begin{pmatrix}2\\3\\2\end{pmatrix}, \overrightarrow{CD}=\begin{pmatrix}1\\m\\1\end{pmatrix}$ **and** $\overrightarrow{AC}=\begin{pmatrix}-2\\-1\\0\end{pmatrix}$ **or** AD $=\begin{pmatrix}-1\\m-1\\1\end{pmatrix}$ **or** BC$=\begin{pmatrix}-4\\-4\\-2\end{pmatrix}$	**B1**	
	$\mathbf{n}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&3&2\\1&m&1\end{vmatrix}=\begin{pmatrix}3-2m\\0\\2m-3\end{pmatrix}$ (so parallel to $\begin{pmatrix}1\\0\\-1\end{pmatrix}$)	**M1A1**	Finds common perpendicular using cross product.
	$\frac{\lvert AC.\boldsymbol{n}\rvert}{\lvert\boldsymbol{n}\rvert}=\frac{\lvert-2(3-2m)+0+0\rvert}{\sqrt{(3-2m)^2+(2m-3)^2}}$ o.e.	**M1**	Uses formula for shortest distance.
	$=\frac{2}{\sqrt{2}}=\sqrt{2}$	**A1**	
		5	
iii	$\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&3&2\\-2&-1&0\end{vmatrix}=\begin{pmatrix}2\\-4\\4\end{pmatrix}\sim\begin{pmatrix}1\\-2\\2\end{pmatrix}$ o.e.	**M1A1**	Finds normal to plane ABC (AEF).
	$\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&3&2\\-1&1&1\end{vmatrix}=\begin{pmatrix}1\\-4\\5\end{pmatrix}$ o.e.	**A1**	Finds normal to ABD (AEF).

	$\cos\theta = \frac{1+8+10}{\sqrt{1^2+2^2+2^2}\sqrt{1^2+4^2+5^2}} \left(= \frac{19}{\sqrt{378}}\right)$	**M1A1FT**	Uses formula for angle between two lines.
	$\Rightarrow \theta = 12.2°$	**A1**	CAO.
		6	

Explanation

10.i

$$\overrightarrow{AB} = \begin{pmatrix}3\\4\\5\end{pmatrix} - \begin{pmatrix}1\\1\\3\end{pmatrix} = \begin{pmatrix}2\\3\\2\end{pmatrix}$$

$$\overrightarrow{CD} = \begin{pmatrix}0\\m\\4\end{pmatrix} - \begin{pmatrix}-1\\0\\3\end{pmatrix} = \begin{pmatrix}1\\m\\1\end{pmatrix}$$

When m=$\frac{3}{2}$

$$\overrightarrow{CD} = \begin{pmatrix}1\\\frac{3}{2}\\1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}2\\3\\2\end{pmatrix} = \frac{1}{2}\overrightarrow{AB}$$ **B1**

So $AB$and$CD$are parallel

ii)

Calculate the common perpendicular of the two lines:

$$\begin{pmatrix}2\\3\\2\end{pmatrix} \times \begin{pmatrix}1\\m\\1\end{pmatrix} = \begin{vmatrix} i & j & k \\ 2 & 3 & 2 \\ 1 & m & 1 \end{vmatrix} = (3-2m)i - (0-0)j + (2m-3)k =$$ **M1A1**

$$(3-2m)\begin{pmatrix}1\\0\\-1\end{pmatrix}$$

Use the formula $\left|\frac{r\cdot n}{|n|}\right|$, where r represents a vector from appoint on AB to a point on CD.

Use r =$\overrightarrow{AC}$

$\overrightarrow{AC} = \begin{pmatrix}-1\\0\\3\end{pmatrix} - \begin{pmatrix}1\\1\\3\end{pmatrix} = \begin{pmatrix}-2\\-1\\0\end{pmatrix}$ **B1**

$\frac{\left|\begin{pmatrix}-2\\-1\\0\end{pmatrix}\cdot\begin{pmatrix}1\\0\\-1\end{pmatrix}\right|}{\sqrt{(1)^2+(0)^2+(-1)^2}} = \frac{|-2|}{\sqrt{2}} = \sqrt{2}$ **M1A1**

iii)

find the normal vector of plane ABC:

$\overrightarrow{AB}\times\overrightarrow{AC} = \begin{pmatrix}2\\3\\2\end{pmatrix}\times\begin{pmatrix}-2\\-1\\0\end{pmatrix} = \begin{vmatrix} i & j & k\\ 2 & 3 & 2\\ -2 & -1 & 0\end{vmatrix}$

$= (0+2)i-(0+4)j+(-2+6)k = \begin{pmatrix}2\\-4\\4\end{pmatrix} = 2\begin{pmatrix}1\\-2\\2\end{pmatrix}$ **M1A1**

m=2

$\overrightarrow{AD} = \begin{pmatrix}0\\2\\4\end{pmatrix} - \begin{pmatrix}1\\1\\3\end{pmatrix} = \begin{pmatrix}-1\\1\\1\end{pmatrix}$

find the normal vector of plane ABD:

$\overrightarrow{AB}\times\overrightarrow{AD} = \begin{pmatrix}2\\3\\2\end{pmatrix}\times\begin{pmatrix}-1\\1\\1\end{pmatrix} = \begin{vmatrix} i & j & k\\ 2 & 3 & 2\\ -1 & 1 & 1\end{vmatrix}$

$= (3-2)i-(2+2)j+(2+3)k = \begin{pmatrix}1\\-4\\5\end{pmatrix}$ **A1**

Find the acute angle between two normal vectors:

$$\cos\theta = \frac{\left|\begin{pmatrix}1\\-2\\2\end{pmatrix}\cdot\begin{pmatrix}1\\-4\\5\end{pmatrix}\right|}{\sqrt{(1)^2+(-2)^2+(2)^2}\sqrt{(1)^2+(-4)^2+(5)^2}} = \frac{|1+8+10|}{\sqrt{9}\sqrt{42}}$$ **M1A1FT**

$$= \frac{19}{3\sqrt{42}}$$

$\theta = cos^{-1}\frac{19}{3\sqrt{42}}$ =12.2° **A1**

The acute angle between two planes are 12.2°

Comment

i. This part of the question was done well, with few slips.

ii. Most candidates remembered the relevant formula for the shortest distance between two lines and applied it accurately. Some assumed that the common perpendicular went through one of the given points. Those who simplified the direction of the common perpendicular to $\begin{pmatrix}1\\0\\-1\end{pmatrix}$ were usually able to come up with the correct numerical answer; others struggled to simplify the expression to eliminate m.

iii. Very few errors were seen in part **(iii)**, with candidates knowing how to find the angle between two planes.

Answers: **(ii)** $\sqrt{2}$ **(iii)** θ = 12.2°

NO. 2

The plane Π_1 has equation

$$\mathbf{r} = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} -4 \\ 1 \\ 3 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}.$$

i. Find a cartesian equation of Π_1. [3]

The plane Π_2 has equation $3x + y - z = 3$.

ii. Find the acute angle between Π_1 and Π_2, giving your answer in degrees. [2]

iii. Find an equation of the line of intersection of Π_1 and Π_2, giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$. [5]

A level FURTHER MATHEMATICS 2018/Winter/Paper1_1/Q8

Mark scheme

Question	Answer	Marks	Guidance
8.i	$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -4 & 1 & 3 \\ 0 & 1 & 2 \end{vmatrix} = -\mathbf{i} + 8\mathbf{j} - 4\mathbf{k}$	**M1**	Finds normal to Π_1.
	$-x + 8y - 4z = 3$	**M1** **A1**	Uses point on plane to find cartesian equation, AEF.
		3	
ii	$\begin{pmatrix} -1 \\ 8 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} = 9\sqrt{11}\cos\theta$	**M1**	Uses scalar product.
	$\Rightarrow \theta = 72.5°$	**A1**	Accept 1.26 rad.
		2	
iii	$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & -1 \\ -1 & 8 & -4 \end{vmatrix} = 4\mathbf{i} + 13\mathbf{j} + 25\mathbf{k}$	**M1** **A1**	Finds direction of line of intersection.
	Point on both planes is, e.g. (1, 1, 1) .	**M1** **A1**	Finds point common to both planes.
	$\mathbf{r} = \mathbf{i} + \mathbf{j} + \mathbf{k} + \lambda(4\mathbf{i} + 13\mathbf{j} + 25\mathbf{k})$	**A1**	States vector equation of line.
		5	

Explanation

8.i

Calculate the normal of the plane:

$$\begin{pmatrix}-4\\1\\3\end{pmatrix}\times\begin{pmatrix}0\\1\\2\end{pmatrix}=\begin{vmatrix}i & j & k\\-4 & 1 & 3\\0 & 1 & 2\end{vmatrix}=(2-3)i-(-8-0)j+(-4-0)k$$

$$=\begin{pmatrix}-1\\8\\-4\end{pmatrix}=-\begin{pmatrix}1\\-8\\4\end{pmatrix}$$ **M1**

Assume the cartesian equation is:

x-8y+4z=d

point 5**i**+**j** is on the plane:

$$5-8\cdot 1+4\cdot 0=d$$ **M1**

d=-3

the equation is

x-8y+4z=-3 **A1**

ii)

the normal of plane l_2 is $\begin{pmatrix}3\\1\\-1\end{pmatrix}$

acute angle between two normal are:

$$cos\,\theta=\frac{\left|\begin{pmatrix}1\\-8\\4\end{pmatrix}\cdot\begin{pmatrix}3\\1\\-1\end{pmatrix}\right|}{\sqrt{(1)^2+(-8)^2+(4)^2}\sqrt{(3)^2+(1)^2+(-1)^2}}=\frac{|3-8-4|}{\sqrt{81}\sqrt{11}}$$ **M1**

$$=\frac{1}{\sqrt{11}}$$

$\theta = cos^{-1}\frac{1}{\sqrt{11}} = 72.5°$ **A1**

So the acute angle between II_1 and II_2 is 72.5°

iii)

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & -1 \\ -1 & 8 & -4 \end{vmatrix} = 4\mathbf{i} + 13\mathbf{j} + 25\mathbf{k}$$ **M1 A1**

Point on both planes is, e.g. (1, 1, 1) . **M1 A1**

r = i + j + k + λ(4i + 13j + 25k) **A1**

Alternative:

solve simultaneous equation:

$3x + y - z = 3$

$x - 8y + 4z = -3$

let z=λ

elimination: $25y - 13\lambda = 12$

$y = \frac{12+13\lambda}{25}$

$x = 8 \cdot \frac{12+13\lambda}{25} - 4\lambda - 3 = \frac{4\lambda+21}{25}$

$$r = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{21}{25} + \frac{4}{25}\lambda \\ \frac{12}{25} + \frac{13}{25}\lambda \\ \lambda \end{pmatrix} = \frac{1}{25}\begin{pmatrix} 21 \\ 12 \\ 0 \end{pmatrix} + \lambda \cdot \frac{1}{25}\begin{pmatrix} 4 \\ 13 \\ 25 \end{pmatrix}$$

Comment

i. Most candidates remembered how to find the Cartesian equation of a plane, recognising the point on the plan as well as the two vectors lying in the plane from the given equation.

ii. Apart from occasional arithmetic errors, most went on to find the angle between the two planes correctly. The most direct and most popular method of finding the line of intersection was by finding its direction using the vector product of the normal, and then working out a point on both lines. Some candidates solved the two equations simultaneously.

Answers: **(i)** $-x + 8y - 4z = 3$ **(ii)** $\theta = 72.5°$ **(iii)** e.g. $\mathbf{r} = \mathbf{i} + \mathbf{j} + \mathbf{k} + \lambda(4\mathbf{i} + 13\mathbf{j} + 25\mathbf{k})$

NO. 3

The plane Π_1 passes through the points (1, 2, 1) and (5, −2, 9) and is parallel to the vector $\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$.

i. Find the cartesian equation of Π_1. [4]

The plane Π_2 contains the lines

$\mathbf{r} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k} + \lambda(\mathbf{i} - 2\mathbf{j} - \mathbf{k})$ and $\mathbf{r} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k} + \mu(2\mathbf{i} + 3\mathbf{j} - \mathbf{k})$

ii. Find the cartesian equation of Π_2. [4]

iii. Find the acute angle between Π_1 and Π_2. [3]

A level FURTHER MATHEMATICS 2017/Summer/Paper1_3/Q9

Mark scheme

Question	Answer	Marks	Guidance
9.i	$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 2 \\ 1 & 2 & 3 \end{vmatrix} = \begin{pmatrix} -7 \\ -1 \\ 3 \end{pmatrix}$	**M1A1**	
	Cartesian equation of Π_1 is $7x + y - 3z = \text{const}$	**M1**	
	$7 \times 1 + 2 - 3 \times 1 = 6 \Rightarrow 7x + y - 3z = 6$	**A1**	
	Total:	**4**	
ii	$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & -1 \\ 2 & 3 & -1 \end{vmatrix} = \begin{pmatrix} 5 \\ -1 \\ 7 \end{pmatrix}$	**M1A1**	
	$5 \times 2 - (-3) + 7 \times 1 = 20 \Rightarrow 5x - y + 7z = 20$	**M1A1**	
	Total:	**4**	
iii	$\cos\theta = \left\lvert \dfrac{\begin{pmatrix} 7 \\ 1 \\ -3 \end{pmatrix} . \begin{pmatrix} 5 \\ -1 \\ 7 \end{pmatrix}}{\sqrt{49+1+9}\sqrt{25+1+49}} \right\rvert$	**M1M1**	
	$\Rightarrow \cos\theta = \dfrac{13}{\sqrt{59}\sqrt{75}} \Rightarrow \theta = 78.7°$ or 1.37 rad.	**A1**	
	Total:	**3**	

Explanation

9.i

Find the normal vector of the plane Π_1

$$\begin{pmatrix}5\\-2\\9\end{pmatrix}-\begin{pmatrix}1\\2\\1\end{pmatrix}=\begin{pmatrix}4\\-4\\8\end{pmatrix}=4\begin{pmatrix}1\\-1\\2\end{pmatrix}$$

$$\begin{pmatrix}1\\-1\\2\end{pmatrix}\times\begin{pmatrix}1\\2\\3\end{pmatrix}=\begin{vmatrix}i & j & k\\1 & -1 & 2\\1 & 2 & 3\end{vmatrix}$$

$$=(-3-4)i-(3-2)j+(2+1)k=\begin{pmatrix}-7\\-1\\3\end{pmatrix}$$

M1A1

Assume the equation is -7x-y+3z=d **M1**

the points (1, 2, 1) is on Π_1

$$-7\times 1-2+3\times 1=d$$

d=-6

$\Rightarrow 7x + y - 3z = 6$. (divided by -1) **A1**

ii.

Find the normal vector of the plane Π_2

$$\begin{pmatrix}1\\-2\\-1\end{pmatrix}\times\begin{pmatrix}2\\3\\-1\end{pmatrix}=\begin{vmatrix}i & j & k\\1 & -2 & -1\\2 & 3 & -1\end{vmatrix}$$

$$=(2+3)i-(-1+2)j+(3+4)k=\begin{pmatrix}5\\-1\\7\end{pmatrix}$$ **M1A1**

Assume the equation is 5x-y+7z=d

the points (2, -3, 1) is on Π_2

5× 2 − (−3) + 7 × 1 = d

d=20

5x-y+7z=20

M1A1

iii.

Find the acute angle between two normal

$$cos\theta = \frac{\left|\begin{pmatrix}-7\\-1\\3\end{pmatrix}\cdot\begin{pmatrix}5\\-1\\7\end{pmatrix}\right|}{\sqrt{(-7)^2+(-1)^2+(3)^2}\sqrt{(5)^2+(-1)^2+(7)^2}}$$

$$= \frac{|-35+1+21|}{\sqrt{59}\sqrt{75}} = \frac{13}{\sqrt{59}\sqrt{75}}$$ **M1M1**

$$\theta = cos^{-1}\frac{13}{\sqrt{59}\sqrt{75}} = 78.7^{\circ}$$

A1

Comment

This question was very well done by candidates, with few errors seen in any part. Candidates were able to calculate vector products correctly to find the normal to the planes and use them to find the equations in **(i)** and **(ii)**. They then used the normal vectors to find the angle between the two planes, and most gave the acute angle as required.

Answers: **(i)** $7x + y - 3z = 6$ **(ii)** $5x - y + 7z = 20$ **(iii)** $\theta = 78.7°$ or 1.37 rad

NO. 4

The plane Π_1 has equation r = –4j – 3k + λ(i – j + k) +μ(i + j – k).

a. Obtain an equation of Π_1 in the form $px + qy + rz = d$. [4]

b. The plane Π_2 has equation **r**.(–5**i** + 3**j** + 5**k**) = 4.
Find a vector equation of the line of intersection of Π_1 and Π_2 . [4]

The line l passes through the point A with position vector a**i** + a**j** + $(a - 7)$**k** and is parallel to $(1–b)$i + b**j**+b**k**, where a and b are positive constants.

c. Given that the perpendicular distance from A to Π_1 is $\sqrt{2}$, find the value of a. [2]

d. Given that the obtuse angle between l and Π_1 is $\frac{3}{4}\pi$, find the exact value of b. [4]

A level FURTHER MATHEMATICS 2023/Summer/Paper1_1/Q7

Mark scheme

Question	Answer	Marks	Guidance
a	$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} \sim \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$	M1 A1	Finds common perpendicular.
	$(-4) + (-3) = -7 \Rightarrow y + z = -7$	M1 A1	Substitutes point.
		4	
b	States point common to both planes e.g $\begin{pmatrix} -7 \\ -2 \\ -5 \end{pmatrix}$.	B1 FT	
	$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 1 \\ -5 & 3 & 5 \end{vmatrix} = \begin{pmatrix} 2 \\ -5 \\ 5 \end{pmatrix}$	M1 A1FT	Finds direction of line.
	$\mathbf{r} = \begin{pmatrix} -7 \\ -2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -5 \\ 5 \end{pmatrix}$	A1	OE.
		4	
c	$\left\lvert \frac{a + a - 7 + 7}{\sqrt{2}} \right\rvert = \sqrt{2}$	M1	Uses correct formula for distance from A to Π_1
	$a = 1$	A1	
		2	
d	$\left\lvert \frac{b + b}{\sqrt{2}\sqrt{(1-b)^2 + 2b^2}} \right\rvert = \frac{1}{2}\sqrt{2}$	M1 A1	Uses correct formula.

	$2b=\sqrt{(1-b)^2+2b^2} \Rightarrow b^2+2b-1=0$	M1	Solves for b.
	$b=-1+\sqrt{2}$	A1	CAO
		4	

Comment

a. This is a standard type of question, and it was generally well-answered. There were a few mistakes in finding the cross-product. Candidates are advised to check carefully as a numerical error at this early stage can affect the remainder of the question.

b. Finding a point common to both planes proved to be quite challenging and many candidates just settled for the marks for finding the line's direction vector. The question asked for the vector **equation** of the line, and it was surprisingly rare to see an answer beginning ***r*** = as required.

c. Quite a few candidates found the foot of the perpendicular from *A* to the plane and used that to write down the distance. This was often successful. Those who did try to use a known formula often omitted the constant term from the equation of the plane.

d. The appropriate formula was more widely known in this part of the question and many candidates were able to apply it correctly and respond confidently to the resulting algebraic challenges. It was common to see the modulus of the length equated to a negative quantity, but this disappeared when the equation was squared. Most remembered to reject the negative solution.

NO. 5

The position vectors of the points A, B, C, D are

$$7\mathbf{i} + 4\mathbf{j} - \mathbf{k}, \quad 11\mathbf{i} + 3\mathbf{j}, \quad 2\mathbf{i} + 6\mathbf{j} + 3\mathbf{k}, \quad 2\mathbf{i} + 7\mathbf{j} + \lambda\mathbf{k}$$

respectively.

a. Given that the shortest distance between the line AB and the line CD is 3, show that $\lambda^2 - 5\lambda + 4 = 0$. [7]

Let Π_1 be the plane ABD when $\lambda = 1$.
Let Π_2 be the plane ABD when $\lambda = 4$.

b. i. Write down an equation of Π_1, giving your answer in the form $\mathbf{r} = \mathbf{a} + s\mathbf{b} + t\mathbf{c}$. [2]

ii. Find an equation of Π_2, giving your answer in the form $ax + + by + cz = d$. [4]

c. Find the acute angle between Π_1 and Π_2. [5]

A level FURTHER MATHEMATICS 2022/Summer/Paper1_3/Q7

Mark scheme

Question	Answer	Marks	Guidance
a	$\overrightarrow{AB} = 4\mathbf{i} - \mathbf{j} + \mathbf{k} \quad \overrightarrow{CD} = \mathbf{j} + (\lambda - 3)\mathbf{k}$	B1	Finds direction vectors of the two lines.
	$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -1 & 1 \\ 0 & 1 & \lambda - 3 \end{vmatrix} = \begin{pmatrix} 2 - \lambda \\ 12 - 4\lambda \\ 4 \end{pmatrix}$	M1 A1	Finds common perpendicular. This may also be done by setting up simultaneous equations and solving them.
	$\left\lvert \frac{1}{\sqrt{(\lambda-2)^2 + (4\lambda-12)^2 + 16}} \begin{pmatrix} -5 \\ 2 \\ 4 \end{pmatrix} \bullet \begin{pmatrix} 2 - \lambda \\ 12 - 4\lambda \\ 4 \end{pmatrix} \right\rvert$	M1 A1	Uses formula for perpendicular distance.
	$\left\lvert \frac{30 - 3\lambda}{\sqrt{17\lambda^2 - 100\lambda + 164}} \right\rvert = 3 \Rightarrow 9(\lambda - 10)^2 = 9\left(17\lambda^2 - 100\lambda + 164\right)$	M1	Sets equal to 3 and forms in quadratic in λ.
	$16\lambda^2 - 80\lambda + 64 = 0$ leading to $\lambda^2 - 5\lambda + 4 = 0$	A1	AG
		7	
b.i	$\mathbf{r} = 7\mathbf{i} + 4\mathbf{j} - \mathbf{k} + s(4\mathbf{i} - \mathbf{j} + \mathbf{k}) + t(-5\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$	M1 A1	OE M1 for using a correct point and attempting to find relevant direction vectors.
		2	
b.ii	$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -1 & 1 \\ -5 & 3 & 5 \end{vmatrix} = \begin{pmatrix} -8 \\ -25 \\ 7 \end{pmatrix}$	M1 A1	Finds normal to the plane Π_2.
	$-8(7) - 25(4) + 7(-1)$ leading to $-8x - 25y + 7z = -163$	M1 A1	Substitutes point.
		4	

c	$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -1 & 1 \\ -5 & 3 & 2 \end{vmatrix} = \begin{pmatrix} -5 \\ -13 \\ 7 \end{pmatrix}$	M1 A1	Finds normal to the plane Π_1.
	$\begin{pmatrix} -5 \\ -13 \\ 7 \end{pmatrix} \bullet \begin{pmatrix} -8 \\ -25 \\ 7 \end{pmatrix} = \sqrt{243}\sqrt{738}\cos\theta$ leading to $\cos\theta = \dfrac{414}{\sqrt{243}\sqrt{738}}$	M1 A1	Uses dot product of normal vectors. cos 0.9776176...
	12.1°	A1	Mark final answer. Accept 0.212°
		5	

Comment

This topic is clearly well understood. Candidates are advised to distinguish carefully between coordinates, position vectors and direction vectors and to use correct notation. Most errors were numerical, particularly in calculating the cross product.

a. This part was generally very well done, and most candidates understood that they needed to show all of their working because they were proving a given answer.

b. i. This was usually successful. The most common error was to omit *r* = from the equation.

ii. This was also well done apart from numerical errors.

c. Despite numerical errors from previous parts it was clear that candidates knew what was required. Very few gave an angle other than the acute one as their final answer.

NO. 6

The lines l_1 and l_2 have equations $\mathbf{r} = -\mathbf{i} - 2\mathbf{j}+\mathbf{k}+s(2\mathbf{i}-3\mathbf{j})$ and $\mathbf{r} = 3\mathbf{i} - 2\mathbf{k} + t(3\mathbf{i}-\mathbf{j}+3\mathbf{k})$ respectively.

The plane Π_1 contains l_1 and the point P with position vector $-2\mathbf{i} - 2\mathbf{j}+4\mathbf{k}$.

a. Find an equation of Π_1, giving your answer in the form $\mathbf{r} = \mathbf{a}+\lambda\mathbf{b} + \mu\mathbf{c}$. [2]

The plane Π_2 contains l_2 and is parallel to l_1.

b. Find an equation of Π_2, giving your answer in the form $ax+by+cz = d$. [4]

c. Find the acute angle between Π_1 and Π_2. [5]

d. The point Q is such that $\overrightarrow{OQ} = -5\overrightarrow{OP}$.
Find the position vector of the foot of the perpendicular from the point Q to Π_2. [4]

A level FURTHER MATHEMATICS 2021/Summer/Paper1_3/Q6

Mark scheme

Question	Answer	Marks	Guidance
a	$-(-2\mathbf{i}-2\mathbf{j}+4\mathbf{k})+(-\mathbf{i}-2\mathbf{j}+\mathbf{k}=\mathbf{i}-3\mathbf{k}$	B1	OE. Finds direction vector from P to a point of l_1.
	$\mathbf{r}=-2\mathbf{i}-2\mathbf{j}+4\mathbf{k}+\lambda(2\mathbf{i}-3\mathbf{j})+\mu(\mathbf{i}-3\mathbf{k})$ $\mathbf{r}=-\mathbf{i}-2\mathbf{j}+\mathbf{k}+\lambda(2\mathbf{i}-3\mathbf{j})+\mu(\mathbf{i}-3\mathbf{k})$	B1 FT	OE. FT *their* $\mathbf{i}-3\mathbf{k}$
		[2]	
b	$\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&-3&0\\3&-1&3\end{vmatrix}=\begin{pmatrix}-9\\-6\\7\end{pmatrix}$	M1 A1	OE. Finds vector perpendicular to Π_2.
	$-9(3)-6(0)+7(-2)=-41$	M1	Uses point on Π_2.
	$9x+6y-7z=41$	A1	
		[4]	
c	$\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&-3&0\\1&0&-3\end{vmatrix}=\begin{pmatrix}9\\6\\3\end{pmatrix}\sim\begin{pmatrix}3\\2\\1\end{pmatrix}$	M1 A1	Finds vector perpendicular to Π_1.
	$\begin{pmatrix}3\\2\\1\end{pmatrix}\bullet\begin{pmatrix}9\\6\\-7\end{pmatrix}=\sqrt{14}\sqrt{166}\cos\alpha$ leading to $\cos\alpha=\dfrac{32}{\sqrt{14}\sqrt{166}}$	DM1 A1 FT	Dot product using their normal vectors.
	48.4°	A1	0.845 rad
		[5]	
d	$\overrightarrow{OF}=\overrightarrow{OQ}+\overrightarrow{QF}=-5\begin{pmatrix}-2\\-2\\4\end{pmatrix}+t\begin{pmatrix}9\\6\\-7\end{pmatrix}=\begin{pmatrix}10+9t\\10+6t\\-20-7t\end{pmatrix}$	M1 A1 FT	Scales $\overrightarrow{OP}$ by a factor of ±5 and uses multiple of their normal to Π_2.

	$9(10+9t)+6(10+6t)-7(-20-7t) = 41$ leading to $290 + 166t = 41$	DM1	Substitutes into the equation of Π_2.
	$t = -\frac{3}{2}$ leading to $\overrightarrow{OF} = \begin{pmatrix} -\frac{7}{2} \\ 1 \\ -\frac{19}{2} \end{pmatrix}$	A1	
		[4]	

Explanation

a)

In the form **r** = **a**+λ**b** + μ**c**, **a** represents position vector of a point on the plane,

b and **c** represents two direction vectors parallel to the plane.

$$l_1 : r = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} + S\begin{pmatrix} 2 \\ -3 \\ 0 \end{pmatrix}$$

2**i**–3**j** is a direction vector and we need one more using P and –**i**– 2**j**+**k**

(B1) $\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} - \begin{pmatrix} -2 \\ -2 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$

(B1) $\Pi_1 : r = \begin{pmatrix} -2 \\ -2 \\ 4 \end{pmatrix} + \lambda\begin{pmatrix} 2 \\ -3 \\ 0 \end{pmatrix} + \mu\begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$

b)

$$l_2 : r = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix} + t\begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix}$$

Calculating the cross product of two direction vector parallel to the plane gives the perpendicular vector of the plane

(M1)(A1)$\begin{pmatrix}2\\-3\\0\end{pmatrix}\times\begin{pmatrix}3\\-1\\3\end{pmatrix}=\begin{pmatrix}i&j&k\\2&-3&0\\3&-1&3\end{pmatrix}$

$=\big((-3)\cdot(3)-0\cdot(-1)\big)i+\big(0\cdot3-2\cdot(3)\big)j+(2\cdot(-1)-3(-3)k$

$=-9i-6j+7k$

In the form $ax+by+cz=d$, a =-9, b = -6, c = 7

Then use point (3,0,-2) to calculate the parameter d:

(M1) $d=3(-9)+0\cdot(-6)+(-2)\cdot7=-41$

(A1) Π_2:$-9x-6y+7z=-41$or multiply by -1 at both sides$9x+6y-7z=41$

c)

Calculate cross product:

(M1)(A1)$\begin{pmatrix}2\\-3\\0\end{pmatrix}\times\begin{pmatrix}1\\0\\-3\end{pmatrix}=\begin{pmatrix}i&j&k\\2&-3&0\\1&0&-3\end{pmatrix}$

$=\big((-3)\cdot(-3)-0\big)i+\big(0\cdot1-2(-3)\big)j+\big(2\cdot0-1\cdot(-3)\big)k$

$=9i+6j+3k=3(3i+2j+k)$

Perpendicular vector of Π_1 is$3i+2j+k$

Angle between two perpendicular vectors:

(M1)(A1)$cos\,\theta=\frac{\begin{pmatrix}3\\2\\1\end{pmatrix}\cdot\begin{pmatrix}9\\6\\-7\end{pmatrix}}{\sqrt{3^2+2^2+1^2}\cdot\sqrt{9^2+6^2+7^2}}=\frac{32}{\sqrt{14}\sqrt{166}}$

(A1)$\theta=cos^{-1}\frac{32}{\sqrt{14}\sqrt{166}}=48.4°$

Acute angle between Π_1 and Π_2 is 48.4° (If θ is larger than 90°, we use 180°-θ)

d)

$$\overrightarrow{OQ} = -5\overrightarrow{OP} = -5\begin{pmatrix}-2\\-2\\4\end{pmatrix} = \begin{pmatrix}10\\10\\-20\end{pmatrix}$$

Let F be the foot of perpendicular.

Find an expression for OF:

QF is perpendicular to Π_2,so it is parallel to 9**i**+6**j**-7**k.**

(M1)(A1)$\overrightarrow{OF} = \overrightarrow{OQ} + \overrightarrow{QF} = \begin{pmatrix}10\\10\\-20\end{pmatrix} + t\begin{pmatrix}9\\6\\-7\end{pmatrix} = \begin{pmatrix}10+9t\\10+6t\\-20-7t\end{pmatrix}$

F is on the plain Π_2. Its position vector satisfies the equation of the plane.

(DM1)$9(10+9t) + 6(10+6t) - 7(-20-7t) = 41$

166t + 290 = 41

$t = -\frac{3}{2}$

(A1) $\overrightarrow{OF} = \begin{pmatrix}10 + & 9\cdot\left(-\frac{3}{2}\right)\\10 + & 6\cdot\left(-\frac{3}{2}\right)\\-20 & -7\left(-\frac{3}{2}\right)\end{pmatrix} = \begin{pmatrix}-\frac{7}{2}\\1\\-\frac{19}{2}\end{pmatrix}$

Comment

In vector questions candidates need to check their work carefully as errors in arithmetic can cause problems as the work progresses. Candidates should make sure they distinguish between the position vector of a point and the direction vector of a line.

a. Many candidates seemed unfamiliar with the parametric vector form of the equation of a plane which was asked for in the question. Several gave the Cartesian equation.

b. This was generally well done with only a few errors in arithmetic

c. Most knew what to do here but errors in arithmetic and sign errors in vector products meant that many did not gain accuracy marks.

d. This was a demanding question for many candidates. Those who adopted the expected approach generally did very well. Some candidates used the vector equation of the plane and two dot products. This was sometimes successful, but a more complex approach led to the greater chance of error. A few made the response more complicated by involving an extra point, often then going on to find the foot of the wrong perpendicular. When introducing an extra point, candidates should define this as, for example, a general point of the plane. Those who worked with the distance of a point from a plane rarely connected this with the requirements of the question to achieve a complete solution.

NO. 7

Let t be a positive constant.

The line l_1 passes through the point with position vector $t\mathbf{i}+\mathbf{j}$ and is parallel to the vector $-2\mathbf{i}-\mathbf{j}$. The line l_2 passes through the point with position vector $\mathbf{j}+t\mathbf{k}$ and is parallel to the vector $-2\mathbf{j}+\mathbf{k}$.

It is given that the shortest distance between the lines l_1 and l_2 is $\sqrt{21}$.

a. Find the value of t. [5]

The plane Π_1 contains l_1 and is parallel to l_2.

b. Write down an equation of Π_1, giving your answer in the form $\mathbf{r} = \mathbf{a}+\lambda\mathbf{b} + \mu\mathbf{c}$. [1]

The plane Π_2 has Cartesian equation $5x - 6y + 7z = 0$.

c. Find the acute angle between l_2 and Π_2. [3]

d. Find the acute angle between Π_1 and Π_2. [3]

A level FURTHER MATHEMATICS 2021/Summer/Paper1_1/Q6

Mark scheme

Question	Answer	Marks	Guidance
a	$\begin{pmatrix}0\\1\\t\end{pmatrix}-\begin{pmatrix}t\\1\\0\end{pmatrix}=t\begin{pmatrix}-1\\0\\1\end{pmatrix}$	B1	Finds vector from any point on l_1 to any point on l_2.
	$\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\-2 & -1 & 0\\0 & -2 & 1\end{vmatrix}=\begin{pmatrix}-1\\2\\4\end{pmatrix}$	M1 A1	Finds common perpendicular.
	$\frac{t}{\sqrt{21}}\left\|\begin{pmatrix}-1\\0\\1\end{pmatrix}\bullet\begin{pmatrix}-1\\2\\4\end{pmatrix}\right\|=\sqrt{21}\Rightarrow t=\frac{21}{5}=4.2$	M1 A1	Uses formula for shortest distance.
		[5]	
b	$\mathbf{r}=\frac{21}{5}\mathbf{i}+\mathbf{j}+\lambda(2\mathbf{i}+\mathbf{j})+\mu(-2\mathbf{j}+\mathbf{k})$	B1 FT	Using their value of t.
		[1]	
c	$\begin{pmatrix}0\\-2\\1\end{pmatrix}\bullet\begin{pmatrix}5\\-6\\7\end{pmatrix}=\sqrt{5}\sqrt{110}\cos\alpha$ leading to $\cos\alpha=\frac{19}{\sqrt{5}\sqrt{110}}$	M1 A1	Uses dot product of −2**j** + **k** and normal.
	Acute angle between l_2 and Π_2 is 90 – α = 54.1°	A1	
		[3]	
d	$\begin{pmatrix}-1\\2\\4\end{pmatrix}\bullet\begin{pmatrix}5\\-6\\7\end{pmatrix}=\sqrt{21}\sqrt{110}\cos\alpha$ leading to $\cos\alpha=\frac{11}{\sqrt{21}\sqrt{110}}$	M1 A1 FT	Dot product using their normal to Π_1.
	76.8°	A1	

		[3]	

Explanation

a) l_1 passes through A(t,1,0), l_2 passes through B(0,1,t)

(B1) $= \begin{pmatrix} 0 \\ 1 \\ t \end{pmatrix} - \begin{pmatrix} t \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -t \\ 0 \\ t \end{pmatrix}$

(M1A1) Use vector product to find a common perpendicular vector

Direction vector of l_1 is $-2\mathbf{i}-\mathbf{j}$, Direction vector of l_2 is $-2\mathbf{j}+\mathbf{k}$

$$\begin{vmatrix} i & j & k \\ -2 & -1 & 0 \\ 0 & -2 & -1 \end{vmatrix} = \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$$

(M1A1) Unit vector of $\begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$ is $\frac{1}{\sqrt{21}} \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$

Scalar product of two vectors above is the shortest length between l_1 and l_2, which equals $\sqrt{21}$

$$\left| \begin{pmatrix} -t \\ 0 \\ t \end{pmatrix} \cdot \frac{1}{\sqrt{21}} \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} \right| = \sqrt{21}$$

$$t + 0 + 4t = 21$$

$$t = \frac{21}{5}$$

b) In the form $\mathbf{r} = \mathbf{a}+\lambda\mathbf{b} + \mu\mathbf{c}$, $\mathbf{b}$ and $\mathbf{c}$ are two vectors parallel to the plane.

Π_1 is parallel to vector $-2\mathbf{i}-\mathbf{j}$ and $2\mathbf{j}+\mathbf{k}$, and point A is on Π_1

(B1) $r = \begin{pmatrix} \frac{21}{5} \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$

c) From the equaion $5x - 6y + 7z = 0$ we know that Perpendicular vector of Π_2 is $5\mathbf{i}-6\mathbf{j}+7\mathbf{k}$.

Direction vector of l_2 is **-2j+k**

Use the defination of scalar product a·b = |a||b|cosθ to calculate the angle:

(M1A1) $\cos\alpha = \frac{\begin{pmatrix}0\\-2\\1\end{pmatrix}\cdot\begin{pmatrix}5\\-6\\7\end{pmatrix}}{\sqrt{5}\cdot\sqrt{5^2+(-6)^2+7^2}} = \frac{12+7}{\sqrt{5}\cdot\sqrt{110}} = \frac{19}{\sqrt{5}\cdot\sqrt{110}}$

Obtain the angle between l_2 and perpendicular vector:

(A1) $\alpha = \cos^{-1}\frac{11}{\sqrt{5}\cdot\sqrt{110}} = 35.89°$

Acute angle between l2 and Π_2 is $90° - \alpha = 90° - 35.89° = 54.1°$

d) Perpendicular vector of Π_1 is **-i+2j+4k** from part a)

(M1A1) $\cos\alpha = \frac{\begin{pmatrix}-1\\2\\4\end{pmatrix}\cdot\begin{pmatrix}5\\-6\\7\end{pmatrix}}{\sqrt{21}\cdot\sqrt{5^2+(-6)^2+7^2}} = \frac{-5-12+28}{\sqrt{21}\cdot\sqrt{110}} = \frac{11}{\sqrt{21}\cdot\sqrt{110}}$

(A1) $\alpha = \cos^{-1}\frac{11}{\sqrt{21}\cdot\sqrt{110}} = 76.8°$

Comment

In general candidates seem comfortable with the work on vectors and know which formula to use.

a. Those who used the formula for the distance of a point from a plane were largely successful, although there were some sign errors in vector products.
   Another method was to use a general point *P* on l_1, a point *Q* on l_2, and the fact that the dot products of the vector *PQ* with the two line directions is zero, to generate linear equations. These could be combined with the given length of *PQ* to find t. This gave scope for introducing errors in algebraic manipulation but showed understanding of the situation.
b. Many candidates seemed unfamiliar with the parametric vector form of the equation of a plane which was asked for in the question.
c. Most candidates knew how to find the angle between the line and the normal to the plane and did so correctly. Many candidates did not then go on to find the angle between the line and the plane, omitting the final step involving subtraction from 90°.

d. This part of the question was often completed accurately.